
Midterm 1 Exam
CIS 5030 Online, Spring 2026

1. For each of the following, give a short answer with justification **A correct answer that does not include a proper justification may not receive any credit.**

- (a) The company “RocketX” manufactures rockets, but the production time is not fixed; it is a non-negative random variable $\mathbf{X}$ representing the time in months. The company claims that the average production time is exactly 1 month (i.e., $E[\mathbf{X}] = 1$). Based on this, a data analyst claims that the average squared production time, $E[\mathbf{X}^2]$, must therefore be at most 1. Is the analyst’s claim “TRUE” or “FALSE”? Provide a justification or a counterexample. **(10 points)**
- (b) In a community of 100 people, a computer system assigns every person a random, unique identification number to anonymize the data. Tom and Mick are two members of this community. They each have exactly 10 friends, and notably, 5 of those friends are mutual (belonging to both friend groups). If we look at the friend with the lowest ID number in Tom’s list and the friend with the lowest ID number in Mick’s list, what is the probability that they are the same person? (Hint: think Min-hash) **(15 points)**

Solution. (a) False: let $\mathbf{X}$ be a random variable which is 0 with probability 1/2 and 2 with probability 1/2. Then $E[\mathbf{X}] = 1$ but $E[\mathbf{X}^2] = 2$.

(b) Let T denote the set of Tom’s friends and M denote the set of Mick’s friends. The minimum ID in T equals the minimum ID in M if and only if the person of minimum ID in $T \cup M$ happens to be in $T \cap M$, which happens with probability $|T \cap M|/|T \cup M| = 5/15 = 1/3$.

2. Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T$ be T independent and identically distributed (i.i.d.) random variables sampled uniformly from the set $[N] = \{1, 2, \dots, N\}$. For any $i \neq j$, the pair $(\mathbf{X}_i, \mathbf{X}_j)$ forms a **1-adjacent collision** whenever the pair satisfies $(\mathbf{X}_i - \mathbf{X}_j) \pmod{N} \in \{-1, 0, 1\}$.

Let E_m be the event that the m -th sample $\mathbf{X}_m$ is **not** a 1-adjacent collision with any of the previous $m - 1$ samples $\mathbf{X}_1, \dots, \mathbf{X}_{m-1}$. In other words, E_m occurs if and only if $(\mathbf{X}_m, \mathbf{X}_j)$ is never a 1-adjacent collision for $j < m$.

Part 1

Suppose that, after $m - 1$ samples $\mathbf{X}_1, \dots, \mathbf{X}_m$, there are no 1-adjacent collisions (i.e., the event $\bigcap_{i=1}^{m-1} E_i$ has occurred). Prove the following upper bound on the probability, over the draw of the m -th sample $\mathbf{X}_m$, that E_m occurs:

$$P(E_m \mid E_1, \dots, E_{m-1}) \leq 1 - \frac{2(m-1)}{N}$$

(10 points)

Part 2

Using the result from Part 1 and the inequalities $1 - x \leq e^{-x}$ and $\sum_{i=1}^T i = T(T+1)/2$, prove that the probability of having **no** 1-adjacent collisions among T samples is bounded by:

$$P(\text{No Collisions}) \leq e^{-\frac{T(T-1)}{N}}$$

(15 points)

Solution. (Part 1) Conditioned on $E_1, \dots, E_{m-1}$, the values $X_1, \dots, X_{m-1}$ are $m-1$ distinct numbers in $[N]$ such that no two are adjacent. X_m leads to a collision if and only if it is placed adjacent to some X_i for $i < m$. The number of unique spots “blocked” by each X_i is at least two (each X_i blocks itself and $X_i + 1 \pmod N$), so there are at least $2(m-1)$ blocked spots in total, meaning the probability that X_m creates a collision is at most $1 - \frac{2(m-1)}{N}$.

(Part 2) The probability there are no collisions can be written as

$$\begin{aligned} \Pr(E_1 \wedge \dots \wedge E_T) &= \Pr(E_1) \cdot \Pr(E_2 \mid E_1) \cdot \dots \cdot \Pr(E_T \mid E_1, \dots, E_{T-1}) \\ &\leq \prod_{t=1}^T \left(1 - \frac{2(t-1)}{N}\right) \\ &\leq \prod_{t=1}^T \exp\left(-\frac{2(t-1)}{N}\right) \\ &= \exp\left(-\frac{2}{N} \cdot \sum_{t=1}^T (t-1)\right) \\ &= \exp\left(-\frac{T(T-1)}{N}\right). \end{aligned}$$

3. A plagiarism detection system executes a randomized comparison between a query document q and all documents x in a database $\mathcal{D}$ containing M documents. When comparing a query document q with a database document x , the following occurs:

- if q plagiarized x , the randomized comparison will always output “Match”.
- if q did not plagiarize x , the randomized comparison will output “Match” in error with probability p . This may create false positives.

The system applies the following two-stage process to generate a list of potential plagiarism matches:

- AND-Stage:** Each document x is compared with q for k independent repetitions. If all k repetitions output “Match”, then the system declares x a *Candidate Match*.
- OR-Stage:** The system executes the **AND-Stage** for L independent repetitions, aggregates all documents which received a *Candidate Match*, and outputs those as the *Final Matches*.

Let $\mathbf{X}$ be the random variable representing the total number of documents which were *Final Matches* for a given query q that did not plagiarize any documents. Derive an expression for the expected value $E[\mathbf{X}]$ in terms of M, p, k , and L .

(20 points)

Solution. Let $X_i \in \{0, 1\}$ be the indicator variable that is set to 1 if the i -th document is a final match. By linearity of expectation, $\mathbb{E}[X] = M \cdot \Pr[X_i = 1]$. Now, observe that $X_i = 0$ if and only if each of the L iterations of the AND phase fail for document i , which happens with probability $1 - p^k$. Therefore, $\Pr(X_i = 0) = (1 - p^k)^L$, giving $\mathbb{E}[X] = M \cdot (1 - (1 - p^k)^L)$

4. For variables $x_1, \dots, x_n$ which may take values in $\{-1, 1\}$ and real numbers a_{ij} for each $i \neq j$, define the function $p(x) = \sum_{i \neq j} a_{ij} x_i x_j$. Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be independent random variables defined by

$$\mathbf{X}_i \in \{-1, 1\} \quad \text{and} \quad \Pr[\mathbf{X}_i = 1] = \Pr[\mathbf{X}_i = -1] = \frac{1}{2}$$

for each $i = 1, \dots, n$ (such $\{\mathbf{X}_i\}$ are called unbiased Rademacher random variables), and define the random variable

$$\mathbf{Z} = p(\mathbf{X}_1, \dots, \mathbf{X}_n) = \sum_{i \neq j} a_{ij} \mathbf{X}_i \mathbf{X}_j.$$

Assume that the numbers a_{ij} satisfy $a_{ij} = a_{ji}$ and the normalization condition $s = \sum_{i \neq j} a_{ij}^2 = 1$.

- (a) Compute the expectation $\mathbb{E}[\mathbf{Z}]$. (5 points)
- (b) Compute the variance $\text{Var}(\mathbf{Z})$. (10 points)
- (c) Using the previous results, apply an appropriate inequality to show that

$$\Pr[|\mathbf{Z}| \geq t] \leq \frac{2}{t^2}$$

for every $t > 0$. (5 points)

Solution. (a) Note that $\mathbb{E}[Z] = \sum_{i \neq j} a_{ij} \mathbb{E}[X_i X_j] = 0$ since X_i and X_j are independent with $\mathbb{E}[X_i] = \mathbb{E}[X_j] = 0$.

- (b) Using (a), $\text{Var}[Z] = \mathbb{E}[Z^2]$.

$$\begin{aligned} \mathbb{E}[Z^2] &= \mathbb{E} \left[\left(\sum_{i \neq j} a_{ij} X_i X_j \right) \cdot \left(\sum_{k \neq \ell} a_{k\ell} X_k X_\ell \right) \right] \\ &= \sum_{i \neq j} \sum_{k \neq \ell} a_{ij} a_{k\ell} \mathbb{E}[X_i X_j X_k X_\ell]. \end{aligned}$$

Observe that whenever (i, j, k, ℓ) contain one index that is distinct from the other three, the expectation $\mathbb{E}[X_i X_j X_k X_\ell] = 0$. Therefore, the only terms that don't vanish in the above expression are those for which $\{i, j\} = \{k, \ell\}$, meaning either $i = k$ and $j = \ell$, or $i = \ell$ and $j = k$. For these terms, $\mathbb{E}[X_i X_j X_k X_\ell] = \mathbb{E}[X_i^2 X_j^2] = 1$. Therefore, we can write

$$\mathbb{E}[Z^2] = \sum_{i \neq j} a_{ij}^2 + a_{ij} a_{ji} = 2 \cdot \sum_{i \neq j} a_{ij}^2 = 2.$$

- (c) This follows directly from (a) and (b) and applying Chebyshev's inequality.